Practical Introduction to High Performance Computing

Parallel Applications on HPC Architectures (Preview: Only Pages 1 to 3)

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1 Introduction

The Fugaku and Leonardo supercomputers are both prominent systems in the current Top500 list, ranking among the world's most powerful supercomputers, but they represent different strengths in HPC.

1.1 Fugaku: Hardware Specifications



Figure 1: Fugaku supercomputer. *Note.* From RIKEN Center for Computational Science (n.d.), Wikimedia Commons.

The Fugaku supercomputer is listed as #7 in the Top500 list. It is located in the Fujitsu RIKEN Center for Computational Science in Kobe, Japan [9]. The system has 158,976 compute nodes, each powered by the Fujitsu A64FX processor based on the Armv8.2-A architecture with 512-bit Scalable Vector Extension (SVE) [5]. The wide vector architecture is an important feature because it facilitates efficient execution of highly vectorisable scientific workloads, such as climate modelling or molecular dynamics. Additionally, the processor includes custom hardware extensions, like hardware barriers, sector cache, and advanced prefetch mechanisms, which all reduce synchronisation overhead and improve memory access efficiency at very large scales. Each node contains 48 computational cores with assistant cores (two in compute nodes and four in I/O nodes), which are dedicated to operating system and communication tasks. The benefit of this separation is that it minimises OS interference and can improve computational efficiency for tightly coupled parallel applications. The processor is manufactured using 7 nm FinFET technology, which contributes to improved energy efficiency compared to earlier large systems [5].

One unique feature of Fugaku is its memory architecture. Each node is equipped with 32 GiB of High Bandwidth Memory (HBM2), which provides approximately 1 TB/s of memory bandwidth. This is significantly higher than traditional DDR-based systems, and it allows the processor to sustain performance on memory-bound workloads. The cache hierarchy consists of 64 KiB L1 data cache per core and an 8 MiB shared L2 cache per core group. These cache sizes are not extreme for a supercomputer, but the architecture prioritises streaming performance and vector throughput over large cache capacity, which reflects a design that is optimised for predictable or pre-determined memory access patterns.

Regarding interconnect, Fugaku uses Tofu Interconnect D, which is a high-dimensional mesh/torus network designed for scalability and low latency [5]. Each node connects through multiple 28 Gbps links across several dimensions, which facilitates efficient communication across the entire system. This network design is effective for applications that rely on MPI (Message Passing Interface) communication, and it allows the supercomputer to scale properly to thousands of nodes. In terms of performance, Fugaku can achieve a theoretical peak performance (Rpeak) exceeding 500 petaflops in double precision. Its measured performance (Rmax) on the High Performance Linpack (HPL) benchmark is lower because of common constraints, such as communication overhead, synchronisation, and memory access inefficiencies. However, Fugaku demonstrates high efficiency (Rmax/Rpeak), which means it has a balanced architecture across compute, memory, and interconnect subsystems [9].

Lastly, Fugaku's storage system is organised hierarchically. The first layer, LLIO, provides fast temporary storage for running jobs. The second layer, FEFS (which is based on the Lustre file system), offers large-scale shared storage for users. A third layer integrates cloud storage for long-term data retention [5]. This multi-tier design can improve I/O throughput and reduce bottlenecks in workloads involving large amounts of data.